

AdventureWorks Sales Report

By Hafsat Sulyman(ALT/SOD/025/4614)

1. Perspective

This project analyses AdventureWorks sales data using Power BI to evaluate overall business performance, product profitability, customer distribution, and sales trends across different regions and time periods. The dashboard was designed to provide management with actionable insights that support data-driven decision-making and business growth.

The analysis focuses on four major areas: Executive Sales Performance, Product Performance, Customer and Territory Analysis, and Sales Trend Analysis.

2. Key Assumptions

- The AdventureWorks dataset is accurate and representative of business operations.
- Revenue is calculated using sales transaction records available in the dataset.
- Blank product categories with no associated sales transactions were excluded from analysis.
- Customer counts are based on unique customer records available in the sales data.
- The lower revenue observed in 2014 is influenced by the dataset ending around June/July 2014 rather than representing a complete annual performance decline.

3. Methodology

The project followed a structured Business Intelligence workflow:

Data Preparation

- Imported seven related AdventureWorks tables into Power BI.
- Cleaned and transformed data using Power Query.
- Removed unnecessary fields such as RowGUID, ModifiedDate and system-generated columns.
- Validated missing values and reviewed data quality issues.

Data Modeling

- Established relationships between Sales, Product, Customer, and Territory tables.
- Created a star-schema-like model to support efficient analysis.
- Validated relationship cardinality and filter propagation.

Data Analysis and Visualization

- Created DAX measures for:

- Total Revenue
- Total Customers
- Total Orders
- Total Quantity Sold
- Products Sold
- Revenue per Customer
- Developed five interactive dashboard pages:
 - Executive Sales Overview
 - Product Performance Analysis
 - Customer & Territory Analysis
 - Sales Trend Analysis
 - Insights & Recommendations
- Implemented slicers, page navigation and interactive visualizations to enable dynamic exploration of the data.

4. Insights and Recommendations

Key Insights

- Total revenue generated was approximately \$109.85 million.
- A total of 31,000 orders were processed with approximately 275,000 units sold.
- Bikes generated the highest revenue at approximately \$95 million, contributing the majority of total sales.
- The Mountain-200 product series emerged as the highest revenue-generating product line.
- North America was the strongest performing region, generating approximately \$79 million in revenue.
- The United States contributed the largest share of country-level revenue at approximately \$63 million.
- Revenue peaked in 2013 at approximately \$44 million before declining in 2014 due to incomplete year data.
- Accessories and Clothing generated significantly lower revenue compared to Bikes and Components.

Recommendations

1. Increase investment in the Bikes product category, as it is the primary revenue driver.
2. Expand successful sales strategies from North America into lower-performing regions.

3. Investigate opportunities to improve sales performance for Accessories and Clothing products.
4. Promote high-performing products through targeted marketing campaigns.
5. Continue monitoring sales trends to identify seasonal demand patterns and improve forecasting.
6. Explore customer retention strategies in regions with lower customer concentration to support future growth.

Conclusion

The AdventureWorks dashboard provides a comprehensive view of sales performance, customer distribution, product profitability, revenue trends and key findings. The findings highlight significant opportunities for growth through product optimization, regional expansion, and improved strategic decision-making.